

Joint Window and Exact-Pressure Optimization for Simple Zeros of the Riemann Zeta Function

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Abstract

We record a computer-assisted research-draft refinement of the positioned-pressure method for simple zeros of the Riemann zeta function on the critical line. The refinement keeps the exact shifted-block pressure accounting from the predecessor work and re-optimizes the seven-term analytic window against the resulting global objective. Subject to the same imported analytic interface and finite-dimensional Gram profile used by the predecessor lineage, and using the local interval certificate described below, the resulting candidate bound is

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T, 2T)}{N(T, 2T)} \geq 0.67329075601922263576 \dots,$$

i.e. 67.3290756019%. This manuscript is a research draft, not a peer-reviewed theorem and not a proof of the Riemann hypothesis.

1 Imported analytic interface

Let $S = N_0^s(T, 2T)$ and $N = N(T, 2T)$. We import the analytic inequality

$$S \geq H_{\text{cert}}N + \Delta(M) - o(N)$$

and the finite- m Gram profile

$$h_m(E) = \begin{cases} E, & 0 \leq E \leq m/(m-1), \\ E/m + 2\sqrt{(m-1)E/m} - 1, & E \geq m/(m-1). \end{cases}$$

These ingredients belong to the Anthropic / `trmdy` / `sxuff` lineage and are not reproved here.

2 Re-optimized window

We use

$$v(s) = \sum_{j=0}^6 c_j \cos(\omega_j s), \quad \omega_0 = \sqrt{2}, \quad \omega_j = 2j\pi \ (1 \leq j \leq 6),$$

with exact rational coefficients

$$(c_0, \dots, c_6) = 10^{-9}(1000000000, 6907835, -9359173, 528441, 1509267, -4923883, 1358707).$$

For this window the computed analytic value is

$$H(v) = 0.6724057024266030290069591819 \dots,$$

and we use the safe rational value

$$H_{\text{cert}} = \frac{6724057}{10^7} = 0.6724057.$$

3 Local seven-point certificate

The pair-weight layout is unchanged from `sxuff/zeta-positioned-pressure`. The pressure coefficients are

$$\frac{1}{2300000000}(831522, 1096590, 1071888, 1071888, 1096590, 831522),$$

so their exact sum is $B = 3/1150$. For the re-optimized window, the interval branch-and-bound computation certifies

$$F(g_1, \dots, g_6) \geq \varepsilon, \quad \varepsilon = \frac{52289}{10^7} = 0.0052289,$$

for all $g_i \geq 0$. The recorded verifier statistics are

$$\text{nodes} = 2334226, \quad \text{pruned} = 1167275, \quad \text{splits} = 1166951,$$

with 1407996 convex boxes, 595297 tangent prunes, and maximum depth 77. Independent reproduction is requested.

4 Exact shifted-pressure averaging

For an m -point block, summing the local seven-point inequality over the $m - 6$ windows gives a pressure term

$$P_B = \sum_{j=1}^{m-1} c_j g_j.$$

If $b_1, \dots, b_6$ are the local pressure coefficients, exact double counting yields

$$\sum_{j=1}^{m-1} c_j = (m-6) \sum_{r=1}^6 b_r = (m-6)B.$$

Across all m shifted partitions, a fixed global gap visits each internal block position exactly once. Thus the shifted average retains the exact factor $(m-6)B$.

Set

$$A_m = \varepsilon(m-6), \quad R_m = h_m(A_m), \quad \eta_m = \frac{R_m}{A_m}.$$

The block defect estimate and exact pressure averaging give

$$\Delta(M) \geq \frac{R_m}{m} S - \frac{\eta_m B(m-6)}{m} N - o(N).$$

Substitution into the imported counting inequality yields

$$\boxed{\frac{S}{N} \geq \frac{mH_{\text{cert}} - \eta_m B(m-6)}{m - R_m} - o(1).}$$

5 Final arithmetic

With $H_{\text{cert}} = 0.6724057$, $\varepsilon = 0.0052289$, and $B = 3/1150$, the integer scan selects $m = 210$. Then

$$A_{210} = 1.0666956,$$

$$R_{210} = 1.0657746255211988006371517667\dots,$$

and

$$\eta_{210} = 0.9991366098455818141906198607\dots$$

Hence

$$\frac{S}{N} \geq 0.6732907560192226357616716519\dots - o(1),$$

and therefore safely

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T, 2T)}{N(T, 2T)} > 0.6732907560.$$

6 Trust boundary and status

The exact pressure-position identity, shifted-partition averaging, window arithmetic, and final high-precision bound are directly checkable from the repository. The local interval certificate has been reproduced in the development environment but should be independently regenerated, ideally with a separate interval implementation. The analytic explicit-formula interface and finite- m Gram profile are imported from prior work. Accordingly, the stated percentage should currently be read as a computer-assisted candidate result rather than an established theorem.

Reproducibility

The repository contains exact parameters in `candidate.json`, arithmetic checks in `src/`, and a GitHub Actions workflow that compiles this manuscript to `paper/main.pdf` whenever the LaTeX source changes.